Clarify the meaning of the following:

- algorithm
- technique
- method
- methodology
- approach
- model

- detection
- inference
- prediction
- forecast

- robust(ness)

Algorithm vs Technique vs Method vs Methodology

Algorithm = a finite, well-defined sequence of steps for solving a problem or producing a result

Technique = a specific procedure or practical way of performing a task

Method = a systematic way of achieving a goal

Methodology = an organized framework of principles governing a set of methods and techniques

Approach = a general way of conceptualizing and addressing a problem, within which different methods, techniques, and algorithms, may be used

Model = a simplified representation of a system, process or phenomenon

ML Model = a parameterized computational representation learned or fitted from data and used to produce outputs for new inputs

Example: Machine Learning

Approach:	machine learning
Methodology:	supervised learning with train/validation/test separation
Method:	artificial neural network
Techniques:	normalization, data augmentation, batch normalization, regularization, oversampling, transfer learning
Algorithm:	backpropagation + gradient-based optimization
ML model:	the trained neural network with learned weights and biases

Detection vs Prediction vs Forecast

Detection = establishing the presence or occurrence of an existing phenomenon from observable evidence.

Inference = deriving a proposition that is not directly observed from propositions or evidence that are available.

Prediction = a statement about a future phenomenon or event before it occurs.

Forecast = a reasoned prediction of a future state or development, derived from available evidence and an explicit or implicit model of its evolution.

Detection is about evidence of presence.

Inference is about deriving what is not directly observed.

Prediction is about future occurrence.

Forecast is about reasoned projection of future development.

Smoke detector

Detection:

sensor signal -> *Smoke is present*

Prediction:

smoke + knowledge -> *A fire will occur*

Forecast:

observations *The fire is expected to reach*
evolution model -> *state X within 20 minutes*

Robust, robustness
Robust system = not unduly sensitive to perturbations

Robustness = the degree to which acceptable behaviour is preserved when assumptions, inputs, or operating conditions are disturbed

Robust programming = a programming style aimed at maintaining correct or acceptable behaviour under invalid input, exceptional conditions, and unexpected events

Robust machine learning algorithm = an algorithm whose performance remains acceptably stable under perturbations, noise, outliers, or moderate deviations from its assumptions

Robust estimator = one whose result is not excessively affected by outliers or modest violations of its assumptions

Reliability vs Fault tolerance vs Resilience vs Robustness

Reliability = the system operates correctly over time under the conditions for which it was designed.

Fault tolerance = the system continues to operate in the presence of specific faults or failures.

Resilience = the system can recover after a disturbance, disruption, or failure.

Robustness = the system's behaviour or results remain acceptable when conditions deviate from the ideal or expected ones.

Fuzziness induces robustness by design: **gradual membership** transitions make the system inherently less sensitive to small perturbations in the input.